ADANI INSTITUTE OF DIGITAL TECHNOLOGY MANAGEMENT

AI in Supply Chain & Logistics

Routing, Demand Forecasting and Warehouse Automation in Indian Logistics

INDIVIDUAL REPORT

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Problem Statement

AI in Supply Chain & Logistics

A major Indian logistics company is dealing with delayed deliveries, high fuel costs, and unpredictable demand. How can AI help optimise routing, predict demand accurately, and automate warehouse operations to improve efficiency and customer satisfaction?

1. Introduction

Logistics links producers, warehouses and customers. When logistics works well prices stay low deliveries arrive on time. Customers feel happy. This report looks at a problem that many large Indian logistics companies face: deliveries that're late fuel costs that are high and demand that is hard to predict. The main question is how artificial intelligence or AI can help make routes better forecast demand accurately and run warehouses automatically. It also asks how much these changes can improve efficiency and make customers happier.

The report has four goals. First it will explain the industry context where this problem happens. Second it will describe the problems in detail. Find out what causes them. Third it will study three AI tools – routing optimization demand forecasting and warehouse automation – using ideas from the course such as search algorithms, machine learning, knowledge representation, reasoning under uncertainty and responsible AI. Fourth it will look at companies – Delhivery, Amazon and Maersk –. Decide how trustworthy and useful that evidence is.

The study uses secondary research. It pulls information from government reviews, rating-agency and industry reports company disclosures and published case studies. No interviews or surveys were done. Because many performance numbers come from vendors or consultants they are treated as hints than facts and the quality of the evidence is discussed in Section 8. Tables, worked examples and other supporting material are in the Appendix. Are cited in the text.

The report is organized as follows. Sections 2 to 4 describe the industry define the problems and analyse their causes. Section 5 looks at the three AI tools. Section 6 compares these tools. Considers how they work together. Sections 7 and 8 talk, about ethics and the limits of the evidence and Section 9 gives the conclusion.

2. Industry Overview

Logistics includes transport, warehousing, inventory control and last-mile delivery. India’s logistics market was worth about USD 240.5 billion in 2025. One industry forecast puts it near USD 800 billion by 2030. The cost of logistics compared to GDP is a topic that is often discussed. Estimates vary because of different definitions. A DPIIT–NCAER assessment placed logistics cost at 7.97 percent of GDP for 2023–24 using data from industry and secondary data from MoSPI, RBI and GSTN. A CII–Knight Frank report puts logistics cost at 10–10.7 percent in FY2026 lower than the 13–14 percent a decade earlier. It notes that India’s Logistics Performance Index rank climbed from 54th in 2014 to 38th in 2023. Numbers of 13–14 percent are still used when compared with the average of 8–9 percent. The differences show that the figure depends on what's counted so it is treated here as a range rather than a single value.

Policy has been a driver of change. The National Logistics Policy and PM GatiShakti aim to integrate transport planning and digital platforms such as the Unified Logistics Interface Platform and the Logistics Data Bank support real-time tracking. Infrastructure such as logistics parks and dedicated freight corridors is intended to move more freight from road to rail; rail’s share was about 27 percent with a target of 40 percent by 2030. So CII–Knight Frank argues that the supply chain has not yet reached optimal efficiency.

The structure of road freight is central to this report. Around 75 percent of truck operators own than five trucks; the sector is largely cash-based and monitoring of driving behaviour, fuel use and maintenance is limited. ICRA similarly reports that small operators account for 75–80 percent of the road logistics market. Large organised players operate differently. At the end of FY2025 Delhivery covered 18,833 PIN codes. Ran 45 fully or semi-automated sortation centres and 111 gateways with a rated automated sort capacity of 8.2 million shipments a day. This contrast between a low-data market and a few highly digitised operators helps explain why AI adoption is uneven and why a large company is the natural setting, for the problem studied here.

3. Problem Definition

Three problems are examined: delayed deliveries, high fuel costs and unpredictable demand. Table A1 in Appendix A summarises each problem and its business impact.

- Delayed deliveries reduce customer satisfaction increase complaints and re-delivery attempts and can trigger penalties or loss of contracts. In logistics service reliability is a part of the product so delayed deliveries are both an operational and a commercial problem.
- High fuel costs are the most financially significant. Fuel accounts for 50–60 per cent of fleet operators' operating costs and other estimates put the figure at about 55 per cent. ICRA expects the operating margins of listed road logistics companies to shrink by up to 200 basis points in FY2027 because of a fuel price shock and freight rates from Delhi to cities were about 7 per cent higher in May 2026 than in February. Costs are passed on slowly because freight rates in an oversupplied competitive market follow demand rather, than cost. CRISIL also reports softening fleet utilisation in 2026, which raises the cost of each trip.
- Unpredictable demand makes planning difficult. Volumes vary by season, festival, region and e-commerce promotion. If demand is under-estimated, vehicles and staff are insufficient. Deliveries are delayed; if it is over-estimated, vehicles and warehouse space sit idle and costs rise.

The problems are linked. Poor demand estimates lead to loads and last-minute capacity, which cause delays and extra trips; extra trips raise fuel use; and high fuel costs squeeze the margins needed to invest in better systems. The research question therefore asks whether AI can address these linked problems through three applications: routing optimisation (delays and fuel) demand forecasting (planning under uncertainty) and warehouse automation (handling speed and accuracy).

4. Root Cause Analysis

Root causes can be grouped into four areas: planning and decision-making visibility and data market structure and warehouse operations. Table A2 links each problem to its causes. The evidence that backs them up. Table A3 displays a "five whys" chain that explains why deliveries get delayed.

- Decision-making: A lot of routes and dispatch plans are still made by hand. Follow set patterns. Static plans do not adjust to traffic, weather, road closures or last-minute changes in orders. Sub-optimal routes add distance and time which makes delays and fuel use rise. Demand planning has the flaw: old forecasting methods use past averages and manual tweaks and they have trouble with seasonality and sudden spikes.
- Data: When fleets do not have telematics operators cannot see driving behaviour, fuel use or maintenance needs. Without data delays are found of foreseen and fuel leaks cannot be measured. Data silos between sales, operations and warehouses stop the view that forecasting requires.
- Market structure: Fragmentation, many trucks and fierce competition keep freight rates linked to demand, not cost. Because of this operators take on fuel price shocks. Cannot pass them on. Low fleet utilisation and empty return trips worsen the issue; fuel is the variable cost and utilisation is often misunderstood as a cost driver.
- Warehouse operations: Manual sorting, put away and picking are slow and full of mistakes. Delhivery's own history shows this: for years after it started last-mile centre staff sorted packages by reading addresses before the company automated how shipments were assigned to delivery drivers. Errors and slow handling at hubs make delays happen before vehicles even leave.

Two points deserve emphasis. First the root causes mainly involve decisions made with late information, which is exactly the type of problem AI can tackle. Second some causes, like diesel prices, road congestion and market oversupply are, beyond the company's control. AI can cut fuel use. The delays caused by bad

planning but it cannot change diesel prices or road conditions. This difference is used in Section 6 to decide how much of each problem AI can realistically address.

5. AI-Based Solutions

AI is the use of computer systems to perform tasks that normally require intelligence. All the applications in this report are examples of AI meaning systems built for one defined task such as finding a route forecasting volume or sorting parcels; general AI, able to reason across any domain and superintelligence remain theoretical. The field has moved through stages. The Turing Test proposed a way to judge machine intelligence rule-based expert systems in the 1980s encoded expertise and AI winters followed when promises exceeded results. The recent rise of machine learning, learning and generative AI has been driven by more data and computing power. Supply chain is one of the business application areas, alongside marketing, human resources, finance and healthcare. The three applications below draw on stages of this evolution and Table C1 in Appendix C links each course concept to its use in logistics.

5.1 Routing optimisation

Problem framing. Routing can be framed as a graph search problem. Hubs, cities and delivery points are nodes roads are edges and each edge has a cost such as distance, time, fuel or tolls. Classical algorithms behave differently on this graph (Table B1 in Appendix B). Breadth- search (BFS) explores level by level and finds the route with the fewest stops but it ignores cost so it can return an expensive route. Depth-first search (DFS) follows one path far as it can before backtracking; it uses little memory but may return a long route and is more useful for checking whether a path exists. Greedy first search, an informed method always expands the node that appears closest to the goal according to a heuristic; it is fast but can be misled. A* search adds the cost already incurred to the remaining cost and returns the lowest-cost route when the estimate never overstates the true cost. In logistics cost can be defined as fuel or time. The heuristic as straight-line distance. Adversarial search, which models competing decision-makers has limited relevance to routing itself although similar reasoning can support pricing or bidding.

Beyond a route. Real fleets must assign vehicles to many deliveries under constraints of capacity time windows and driver availability. This is the vehicle routing problem, a business decision problem solved with optimisation and heuristics rather than a single search. AI route planning weighs traffic, vehicle capacity, driver availability, delivery priorities and customer time windows and updates plans as conditions change. Reinforcement learning, in which a system learns from rewards such as fuel use or time can adapt routes further as conditions evolve.

Evidence. Delhivery replaced manual shipment allocation at its distribution centres with an allocation system built on the Gurobi optimisation solver and later reports describe in-house AI prediction systems across its network. Maersk uses twins to simulate voyages and forecast fuel use, emissions and routing efficiency and applies optimisation and reinforcement learning to select speed and route. Its reported fuel savings range from 9.2 per cent to 12 per cent and the sources conflict. In road transport McKinsey Global Institute work reported that a European trucking company cut fuel costs by 15 per cent through AI routing and UPS's ORION system is reported to cut about 100 million miles of truck travel a year. A vendor case study reports 8–20 per cent fuel use and 15–25 per cent better on-time delivery.

Assessment. Routing addresses two of the three problems at once because shorter and better-timed routes reduce both lateness and fuel burned. Its benefits however depend on location and traffic data, driver acceptance and the ability to follow recommended routes on congested toll-heavy Indian roads. Sea routing at Maersk and parcel routing at UPS differ from Indian trucking so the percentages should be read as indicative ranges.

5.2 Demand forecasting

Techniques. Forecasting is mainly a learning task: models learn from labelled historical data, such as past shipments by lane, date, customer, festival and weather to predict future volumes. Unsupervised learning supports the task by clustering customers or zones and by flagging anomalies, such as a sudden spike on one lane. Deep learning, recurrent networks such as LSTMs can capture seasonality and non-linear patterns that simple averages miss for example a festival surge on top of a weekly cycle. Forecast quality is typically measured with error metrics such as the absolute percentage error (MAPE); Table C2 in Appendix C shows a worked example in which forecasts of 90, 130 and 90 against actual volumes of 100, 120 and 80 give a MAPE of about 10.3 per cent.

Demand can never be known with certainty useful forecasts express uncertainty. Probabilistic reasoning, for example Bayesian methods produces statements such as a 70 per cent chance that volume exceeds a given level, which can guide how capacity to reserve or how many vehicles to position in advance. Generative AI adds an interface layer: a planner can ask in language why volume is rising in a region and receive a summary of the model's drivers, which supports transparency.

Evidence. Amazon's regionalisation strategy depends on predicting what will be bought where so inventory can be placed near customers; the company reported that more than 74 per cent of US orders came from fulfilment centres within the customer's region. Its newer forecasting model adds external data such as weather and regional preferences to sales history and a trade report cites a 20 per cent improvement in forecasts although this is not independently verified. Industry sources attribute to McKinsey a reduction in forecast error of 20–50 per cent and up to 65 per cent stockouts from AI forecasting and a reduction of 20–30 per cent in inventory in distribution operations. These figures are repeated widely by vendors. Should be treated cautiously.

Assessment. Better forecasts do not directly shorten a route. They improve the plan around it: vehicles and staff can be positioned before demand arrives loads can be consolidated and warehouse space used more efficiently which reduces both delay and idle capacity. The main constraint is data. Forecasting needs a history of clean integrated data and models can inherit bias for example by under-predicting regions that were historically under-served. A Gartner survey reported in August 2026 found that 55 per cent of supply chain officers were unclear on the return on their AI investments a reminder that value depends on execution.

5.3 Warehouse automation

Techniques. Warehouse automation brings together parts of the syllabus. Rule-based and expert systems encode decisions explicitly for example "if an item is perishable pick it first". If a zone exceeds 90 per cent capacity redirect inbound stock". Rules are transparent and easy to audit though they become hard to maintain as conditions multiply. Knowledge graphs and ontologies represent SKUs, bins, vehicles orders and customers as entities so a system can reason across relationships, such as finding the nearest suitable vehicle for a delayed shipment or storing frequently co-ordered items together. An ontology provides a shared vocabulary so that data from systems carries the same meaning, which matters when business data is spread across warehouse, transport and sales systems. Deep learning supports computer vision for reading labels sorting parcels and detecting damage and reinforcement learning can coordinate fleets of picking or sorting robots.

Evidence. Delhivery's automated sortation network is the Indian example: 45 fully or semi-automated sortation centres and 111 gateways with rated automated sort capacity of 8.2 million shipments a day at March 2025 up from 3.7 million in December 2021. Amazon reported in 2023 that 75 per cent of customer orders were handled partly by robotics, a capability that traces back to its 2012 purchase of Kiva Systems and it is developing robots that respond to natural-language commands. In India Flipkart's robotic sortation pilot reported a 60 per cent gain in process efficiency per hundred vehicles and GreyOrange robots were reported to complete in an hour work that takes an average worker five.

Assessment. The economic evidence is mixed. GreyOrange estimated payback at 18 months whereas an older industry article reported 8–10 years in India against 2–3 years in Japan because labour is cheap. Automation therefore appears justified where volumes are high and labour constraints or error costs are significant as in large sortation hubs. There is also a workforce dimension: a GreyOrange presentation suggested its systems could replace 60–80 per cent of a warehouse workforce while Amazon describes its approach as robotics that

changes jobs rather than removing them. Both claims come from organisations with an interest, in the technology.

6. Integrated View and Comparative Evaluation

The three applications are most valuable when they are connected. Forecasts show how much volume will arrive, where it will arrive and when it will arrive. Warehouse systems then use that information to plan staff and stock. Routing then assigns vehicles and orders deliveries. The actual delivery outcomes are fed back as training data for the forecasting models. This loop cuts delays, fuel waste and idle capacity at once. That is why large operators choose to integrate forecasting, execution and delivery of keeping them separate.

The applications can be compared on four criteria: their effect on the three problems, how capital they need the time to see benefits and how much they depend on data and execution. This comparison comes from the author's judgement based on the evidence not from a measured result. Routing appears strongest because it tackles delays and fuel directly relies mostly on software and telematics and can be tested on a corridors. Forecasting has a more indirect effect and takes longer because the models need historical data to calibrate. Warehouse automation has the capital cost and the least certain payback in a low-wage market but its case becomes stronger at high-volume hubs. The cases mostly follow this pattern: software-led capability usually comes before automation.

The cases also show limits to transfer. Amazon and Maersk run at large scale and use a lot of capital. Maersk's routing deals with sea, not congested roads. Amazon's robotics economics reflect labour costs. Delhivery, a network operator is the closest comparator. A simple sensitivity calculation (Table D1) shows the scale of gains. If fuel is 55 percent of operating cost a fuel saving of 8 to 12 percent equals roughly 4.4 to 6.6 percent of operating cost. This range is cautious compared with the 15 percent reported in the trucking case and is based on assumptions rather, than company data.

7. Ethical, Legal and Responsible AI Considerations

Responsible AI covers bias, transparency, privacy and regulation and each applies to these systems.

- Bias: Models learn from data so patterns of past under-service can be reproduced. A forecasting model trained on years in which a region received services may predict low demand there which then justifies continued low service. Regular audits of forecasts and outcomes by region are one safeguard.
- Accountability: Planners and drivers are more likely to trust and correctly use a route or forecast if they can see why it was produced. Rule-based components and explainable outputs. Human override should remain possible particularly when an automated decision affects a delivery commitment or a worker's schedule.
- Privacy: Gps track driver location and behaviour. India's Digital Personal Data Protection Act 2023 sets obligations for processing data so companies need a clear purpose, consent where required and limits, on retention. Monitoring that improves safety and fuel efficiency can become surveillance if it is used to penalise drivers without context.
- Workforce impact. Automation changes the composition of warehouse and sorting jobs. Reskilling and gradual deployment reduce harm and claims that robots will replace most of a workforce should be tested against conditions, where low wages already weaken the cost case.
- Regulation: Systems can fail when data is conditions change so model performance needs continuous monitoring. Regulation of AI is still developing, so organisations must meet existing data, labour and safety laws while preparing for requirements.

8. Limitations of the Evidence

This report uses information from other people or organizations and there are some things that make it hard to fully trust the results. First a lot of the numbers come from company blogs reports from experts or repeated mentions of McKinsey and Gartner but often without the research behind them so the percentages given are just a rough idea. Second the numbers don’t always match up: one place says that Maersk saved 9.2 per cent on fuel and that saves than 300 million dollars a year while another says it saved 12 per cent and 150 million dollars a year. Also the cost of logistics compared to the size of the economy ranges from 7.97 per cent to 14 per cent depending on how it's measured and when the data was collected. Third the numbers for Amazon are mostly from the company itself. Some of them are from 2023. For Delhivery there is information about their buildings and equipment but not much about how their AI is working. Fourth some of the information about warehouse robots like from Flipkart and GreyOrange is years old and might not match what is happening now with robot prices. Fifth the company, in the example is not real. There is no real data to check the results against. In the future getting data like talking to people who manage logistics or testing small projects could help make the numbers given here more accurate.

9. Conclusion

This report looked at how artificial intelligence might help an Indian logistics company that was struggling with late deliveries, rising fuel costs and demand that was hard to predict. The study found that all three issues had one thing in common: decisions were being made based on incomplete or scattered information. The market’s structure and fuel prices also limit what a single company can actually control.

Under these conditions the report examined three AI solutions. First routing optimisation uses graph search, optimisation methods and reinforcement learning to reduce both delivery delays and fuel consumption. Demand forecasting uses supervised learning, unsupervised learning, deep learning and probabilistic reasoning to make better plans when the future is uncertain. Third warehouse automation uses rule-based systems, knowledge graphs and robotics to speed up and improve handling in high-volume settings.

Real-world examples from companies like Delhivery, Amazon and Maersk show that AI can deliver results in each of these areas.. The data often comes from the companies themselves or from secondary sources. Also the financial impact in India may differ, especially when automation requires upfront investments.

Using AI responsibly means paying attention to fairness, transparency, data privacy and how it affects workers. In the end the evidence shows AI can lower fuel use, cut delays and reduce uncertainty in planning.. It only helps with problems caused by poor decision-making. It cannot fix issues, like fuel prices or road conditions. To know for sure how benefit AI can bring, primary research and company-specific pilots would be needed.

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Appendix

Appendix A: Problems and root causes

Table A1. Problems and business impact

#	Problem	Business impact
1	Delayed deliveries	Lower customer satisfaction, penalties, lost contracts
2	High fuel costs	Margin squeeze; fuel is about 50–60% of fleet operating cost
3	Unpredictable demand	Idle capacity in slack periods, shortages at peaks

Table A2. Root causes by problem

Problem	Root causes	Supporting evidence
Delays	Manual, static route planning; poor visibility of fleet and driver behaviour; congestion; slow, error-prone warehouse handling	Little oversight of driving behaviour, fuel use and maintenance
High fuel cost	Sub-optimal routes and idling; low utilisation and empty return trips; exposure to diesel prices with weak pass-through	Freight rates follow demand, not fuel cost; fuel is the highest variable cost
Unpredictable demand	Forecasts based on historical averages and manual adjustment; siloed data; seasonal and e-commerce spikes	Traditional methods rely on historical averages

Table A3. “Five whys” chain for delayed deliveries (illustrative)

Why?	Answer
1. Why was the delivery late?	The vehicle left the hub after its scheduled time and took a slow route
2. Why did it leave late?	Parcels were sorted and loaded manually and loading ran over time
3. Why did loading run over?	Volume at the hub was higher than planned
4. Why was volume higher than planned?	The plan was based on historical averages, not current demand signals
5. Why were current signals not used?	Data from sales, operations and warehouses was not integrated

Appendix B: Search algorithms

Table B1. Search algorithms applied to routing

Algorithm	Type	How it chooses the next node	Strength	Weakness in routing
Breadth-first search (BFS)	Uninformed	Level by level	Fewest stops	Ignores distance, fuel and traffic
Depth-first search (DFS)	Uninformed	Follows one path deeply	Low memory	May return a long, costly route
Greedy best-first search	Informed	Lowest heuristic estimate to goal	Fast	Can miss the cheapest route
A* search	Informed	Lowest cost so far plus estimate to goal	Finds lowest-cost route if estimate is never too high	Needs a good cost and heuristic definition

Appendix C: Course concepts and forecasting accuracy

Table C1. Course concepts applied to logistics

Course concept	Application area	Use in logistics
Narrow AI	All	Each system performs one task (routing, forecasting, picking)
BFS, DFS, greedy search, A*	Routing	Finding low-cost routes on a road network graph
Optimisation (vehicle routing)	Routing	Assigning vehicles to deliveries under constraints
Reinforcement learning	Routing, warehouse	Adapting routes and coordinating robots through rewards
Supervised learning	Forecasting	Predicting volumes and arrival times from labelled history
Unsupervised learning	Forecasting	Clustering customers or zones; detecting anomalies
Deep learning (LSTM, CNN)	Forecasting, warehouse	Seasonal patterns; parcel sorting and damage detection
Probabilistic reasoning	Forecasting	Expressing demand as ranges with probabilities
Rule-based and expert systems	Warehouse	Transparent picking and slotting rules
Knowledge graphs and ontologies	Warehouse	Linking SKUs, bins, vehicles and orders for reasoning
Generative AI	Forecasting, customer service	Plain-language planner assistants and customer updates
Bias, transparency, privacy, regulation	All	Audits, explainability, driver data protection, compliance

Table C2. MAPE worked example

Week	Actual shipments	Forecast	Absolute error	Percentage error
1	100	90	10	10.0%
2	120	130	10	8.3%
3	80	90	10	12.5%
MAPE (mean)				10.3%

Appendix D: Sensitivity analysis

Table D1. Illustrative fuel-saving sensitivity (assumed figures, not company data)

Fuel saving from routing	Annual saving (₹ crore)	Share of operating cost
5%	5	2.7%
8%	8	4.4%
12%	12	6.6%
15% (European trucking case)	15	8.2%

Assumptions: annual fuel spend of ₹100 crore; fuel about 55% of operating cost, so total operating cost is about ₹182 crore.